

Cache Memory

Cache Memory Mapping Technique

- Cache Memory mapping technique means how data is copied/mapped from MM to CM
 - Mapping Technique:
 - **Direct Mapping:**
 - In this mapping, MM blocks are copied to a fixed block of CM, but once at a time
 - Let, CM block no.= C_b
 - MM block no. = M_b
 - Number of blocks in CM = C
 - Number of blocks in MM = M
 - CM block no. = (MM block no.)mod (Number of blocks in CM)
- $$C_b = M_b \bmod C$$

Cache Memory

- Suppose,
- Main memory size = $512 * 8$ [512 bytes means it contains 512 words and each word is 8 bit]
- Cache memory size = $64 * 8$ [64 bytes means it contains 64 words and each word is 8 bit]
- Block size = 4 words
- So, number of MM blocks = $(\text{Total MM words}) / \text{Block size} = 512 / 4 = 128$ blocks
- Number of CM blocks = $(\text{Total CM words}) / \text{Block size} = 64 / 4 = 16$ blocks
- So, CM block no. = $0 \bmod 16 = 0$
- $16 \bmod 16 = 0$ and so on

Cache Memory

- Number of MM blocks in one block of CM = (Number of MM blocks)/(Number of CM blocks)
- = $128/16 = 8$
- Block size = 4 words
- If there are 512 words in MM then there will be 512 locations
- Thus, MM address = $512 = 2^9$ bits
- So, Each MM address will be 9 bits
- **MM address is divided into 3 parts:**

Tag bit =3	Block bit=4	Word bit=2
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- Index bit = Block bit + Word bit
- **Tag bit** indicates Number of MM blocks in each CM blocks
- **Block bit** indicates Number of CM blocks

Cache Memory

- **Associative mapping in Cache Memory:**

- In associative mapping, MM blocks are copied into any block of CM
- Let, MM size is = 512×8 (512 words and 8 word size)
- CM size is = 64×8 (64 words and 8 word size)
- Block size = 4
- So, Number of MM blocks = $512/4 = 128$ blocks
- Number of CM blocks = $64/4 = 16$
- Thus, number of MM blocks in one block of CM = $128 = 2^7$
- MM size = $512 = 2^9$; Thus, MM address bits = 9
- **Main Memory address format:**

Tag bit = 7

Word bit = 2

Cache Memory

- **Set Associative Mapping:**

- It is a combination of **direct mapping and associative mapping**
- Set Associative Mapping = Direct Mapping + Associative Mapping
- In this method, CM is divided into sets:
- Set = Group of Blocks
- Block = Group of Words
- $\text{CM set no.} = (\text{MM block no.}) \bmod (\text{Number of sets in CM})$
- $\text{Number of sets in CM} = (\text{Number of CM blocks}) / \text{Set size}$
- $\text{Number of blocks in CM} = (\text{Number of words in CM}) / \text{Block size}$
- $\text{Number of blocks in MM} = (\text{Number of words in MM}) / \text{Block size}$

Cache Memory

- Let Set size=2 (Two way set associative)
- Block size = 4 words
- Main memory size 512x8
- CM size =64x8
- Number of MM blocks = $512/4=128$ blocks
- Number of CM blocks = $64/4 =16$ blocks
- Number of sets in CM = $16/2 = 8$ sets = 2^3
- CM set number:

0 mod 8=0

8 mod 8=0

16 mod 8=0

1 mod 8=1

9 mod 8=1

17 mod 8=1

2 mod 8=2

10 mod 8=2

18 mod 8=2

Cache Memory(Set associative mapping)

Set 0	Tag bit	B ₀	Tag bit	B ₈
Set 1	Tag bit	B ₁	Tag bit	B ₉
Set 2	Tag bit	B ₂	Tag bit	B ₁₀
Set 3	Tag bit	B ₃	Tag bit	B ₁₁
Set 4	Tag bit	B ₄	Tag bit	B ₁₂
Set 5	Tag bit	B ₅	Tag bit	B ₁₃
Set 6	Tag bit	B ₆	Tag bit	B ₁₄
Set 7	Tag bit	B ₇	Tag bit	B ₁₅

Cache Memory: 64x8

Block 0
Block 1
Block 2
Block 3
Block 4
.....
Block 8
Block 9
.....
.....
.....
Block 126
Block 127

Main Memory: 512x8

Cache Memory

- Number of MM blocks in one set of CM = (Number of MM blocks)/(Number of CM Sets)
(2 at a time) $= 128/8 = 16 = 2^4$

Main Memory Address format:

Tag = 4 bit

Set = 3 bit

Word = 2 bit